BUSN 32120 · FINAL PROJECT

Which economic environment is better for a beginner investor?

How Fed policy, inflation, recessions, and S&P 500 returns shape the choice between cash and stocks

Data FRED macro · S&P 500 · NBER recessions | **Tools** Python · SQL

Team 5 · Spring 2026

1 THE QUESTION

A high savings rate is not automatically a good return

Our question

When are cash-like returns attractive, and when does the stock market look more favorable?

The nominal rate alone does not tell the full story. The key concept is the real rate: the nominal rate minus inflation.

WHO IT'S FOR

Beginner investors and finance students deciding whether to hold more cash or invest in the S&P 500

WHY IT MATTERS

Real rate = nominal rate − inflation

If inflation is higher than the cash return, then the investor is still ***losing purchasing power*** — even when the headline rate looks good.

Three datasets, two languages, one long macro history

FRED MACRO

Fed Funds Rate, CPI inflation, unemployment

S&P 500

Monthly and 12-month stock returns

RECESSION INDICATOR

Compare expansions and recessions

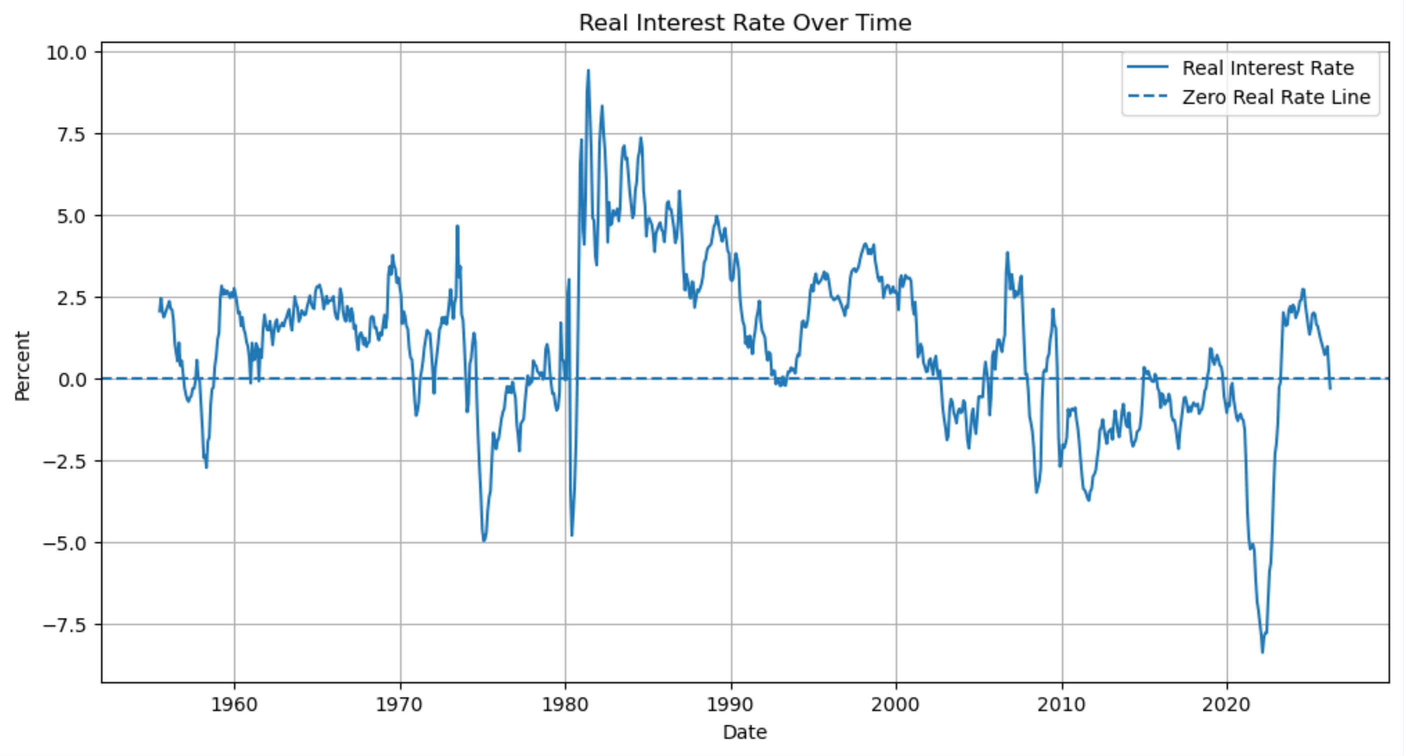
PYTHON
cleaning · feature engineering · EDA · charts · 2 models

SQL
joins · window functions · group by · subqueries

SCOPE:

The macro part covers the full sample: **861 monthly observations, 1954–2026.** **The S&P 500 part** only covers the months where index data is available — **119 months** because the long-history S&P 500 download failed and the notebook fell back to FRED.

Real rates matter much more than nominal rates



From our notebook — Chart 2: real interest rate over time (dashed = zero line)

33%

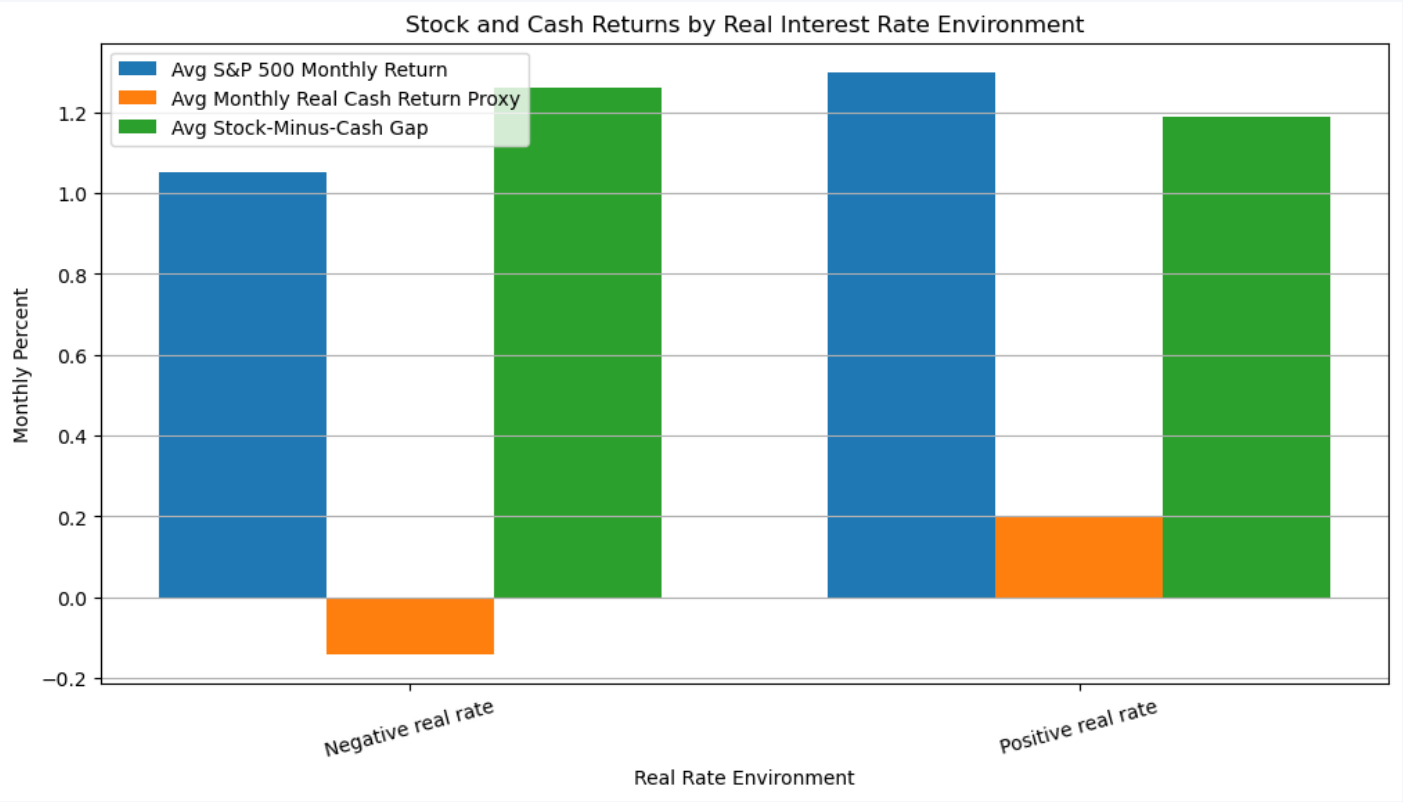
of usable months (283 of 849) had a negative real rate — cash was losing purchasing power in real terms

So what?

Real rates were deeply negative in the mid-1970s and again around 2022; the early 1980s had very high positive real rates.

4 FINDING 2 — ENVIRONMENTS

Cash becomes much more defensible when the real rate is positive



From our notebook — Chart 7: stock & cash returns by real-rate environment

Reading the chart

When real rates are negative, the cash proxy loses about 0.14%/mo. When real rates are positive, cash earns about 0.20%/mo — it is actually protecting purchasing power.

In our available S&P 500 sample, stocks are positive in both environments and higher than the cash proxy — but the S&P sample is shorter, so this evidence cannot be used as a rule.

Takeaway: whether cash is good or bad depends on the real-rate environment.

5 MODELS

Macro is weak for prediction, but is a good indicator of the environment

MODEL 1 · LINEAR REGRESSION

Explain monthly S&P 500 returns from macro variables

$R^2 \approx -0.77$

A test R^2 of about −0.77 means the model does **worse than predicting the average return**. Monthly returns are noisy and macro isn't a good short-term trading signal. The R^2 value is showing the limit of macro for forecasting monthly equity returns.

MODEL 2 · LOGISTIC REGRESSION

Classify high-inflation months (YoY inflation > 3%)

ROC AUC 0.80

Accuracy 0.68 · Precision 0.60 · Recall 0.67 · F1 0.63

Falling unemployment and increasing rate are useful signals for high-inflation environments, which lead to tighter labor markets.

6 TAKEAWAYS

Real beats nominal; macro is context, not a trading signal; know your sample

Real beats nominal

Cash is truly attractive when the return clears inflation. About one in three months had a negative real rate.

Context, not signal

Macro indicators help investors understand the environment, but they don’t reliably predict monthly S&P 500 returns.

Know your sample

Macro covers a long history, 1954–2026; the S&P 500 analysis in this run is the last decade of available data.

NEXT Add T-bill yields, bond ETF returns, and consumer sentiment; backfill a true long-history S&P 500 source, which would make the market return analysis stronger.